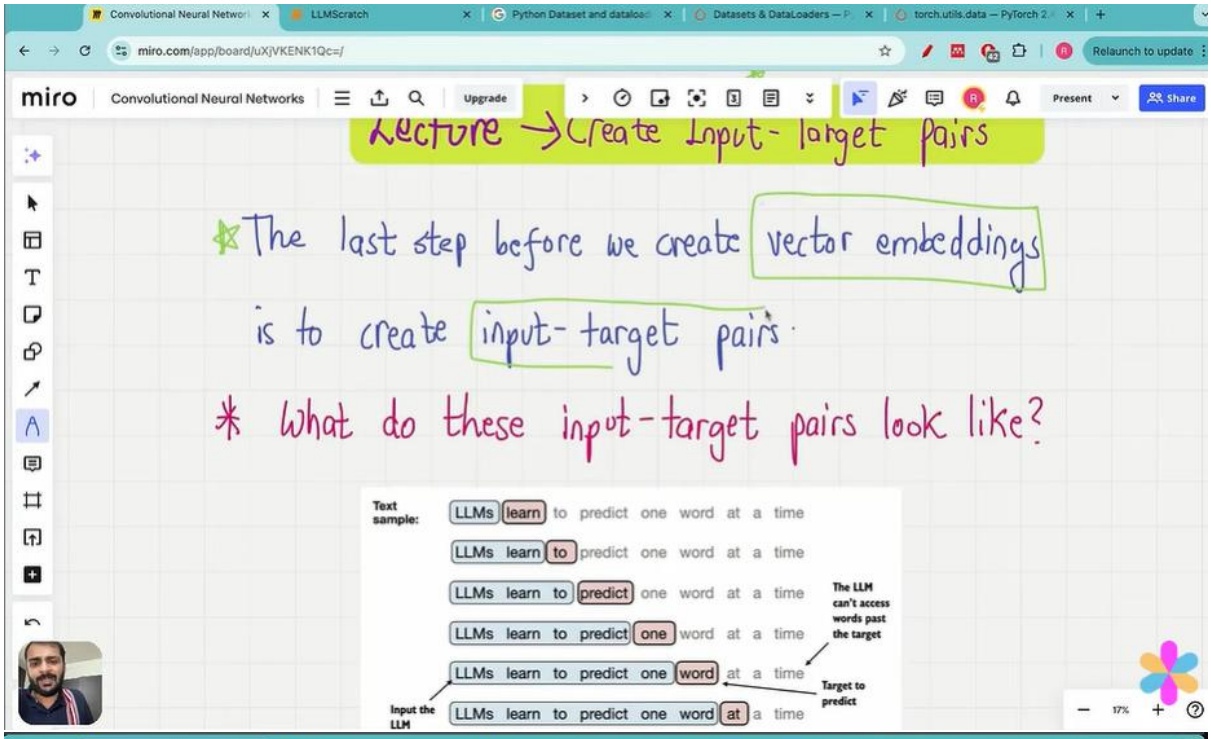
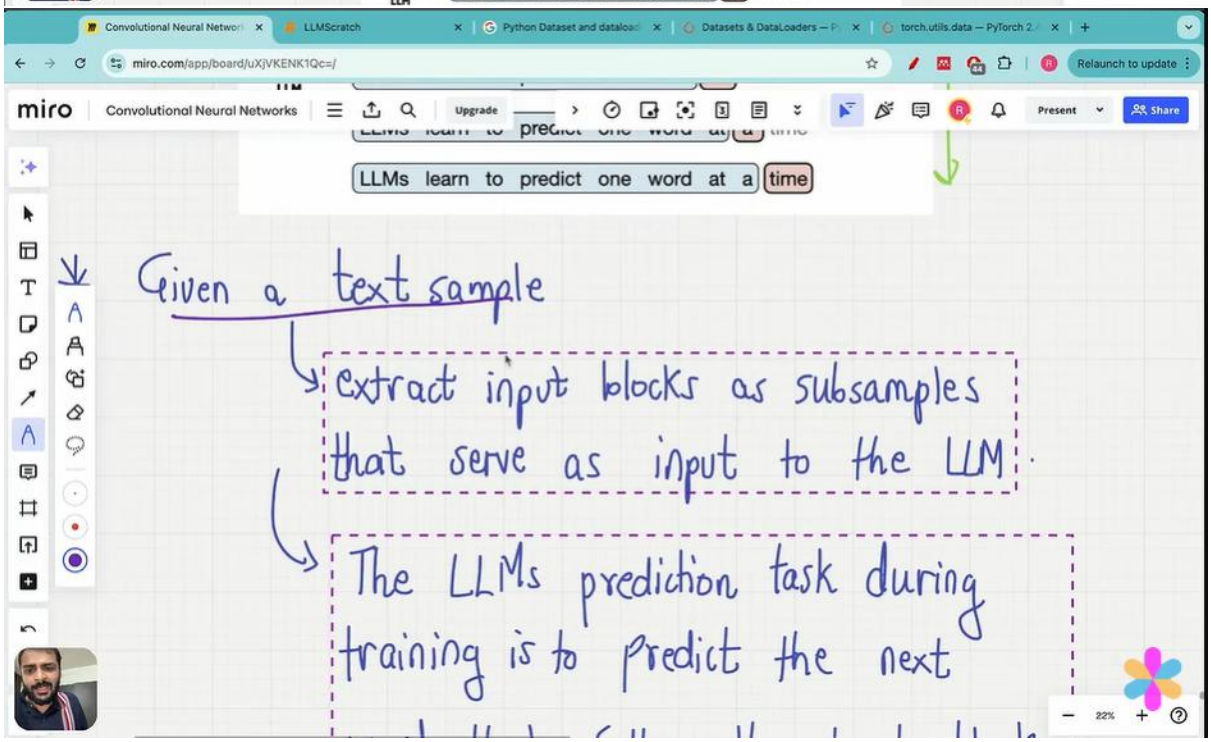
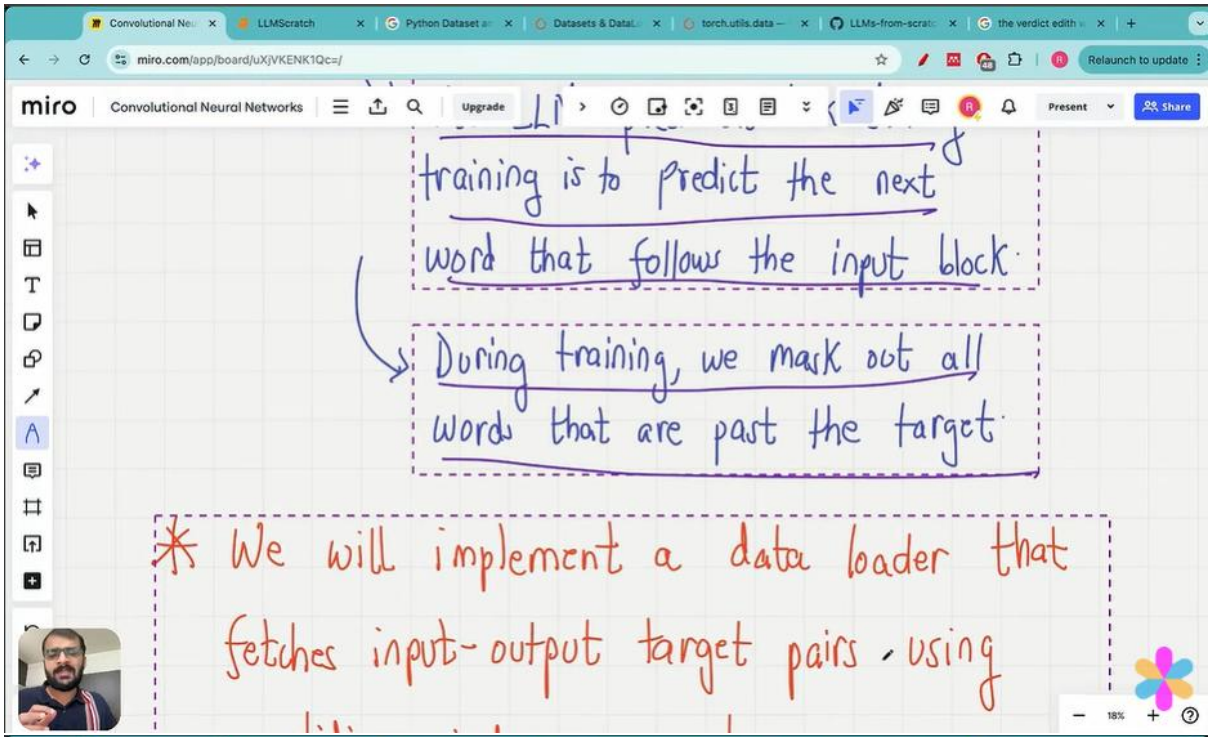
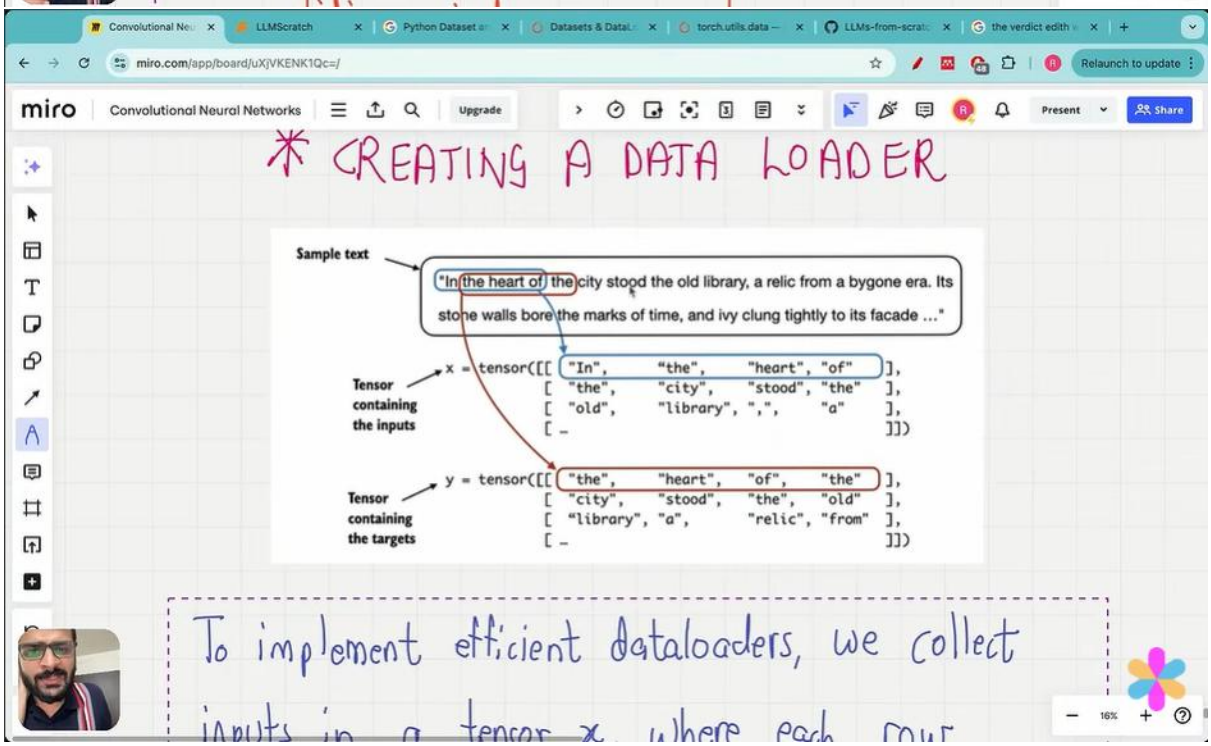
Lec 09


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Convolutional Neural Networks
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Present
Share
Lecture → Create Input-Target pairs
The last step before we create vector embeddings is to create input-target pairs.
* What do these input-target pairs look like?
Text sample:
LLMs learn to predict one word at a time
LLMs learn to predict one word at a time
LLMs learn to predict one word at a time
LLMs learn to predict one word at a time
LLMs learn to predict one word at a time
LLMs learn to predict one word at a time
The LLM can't access words past the target
Target to predict
Input the LLM

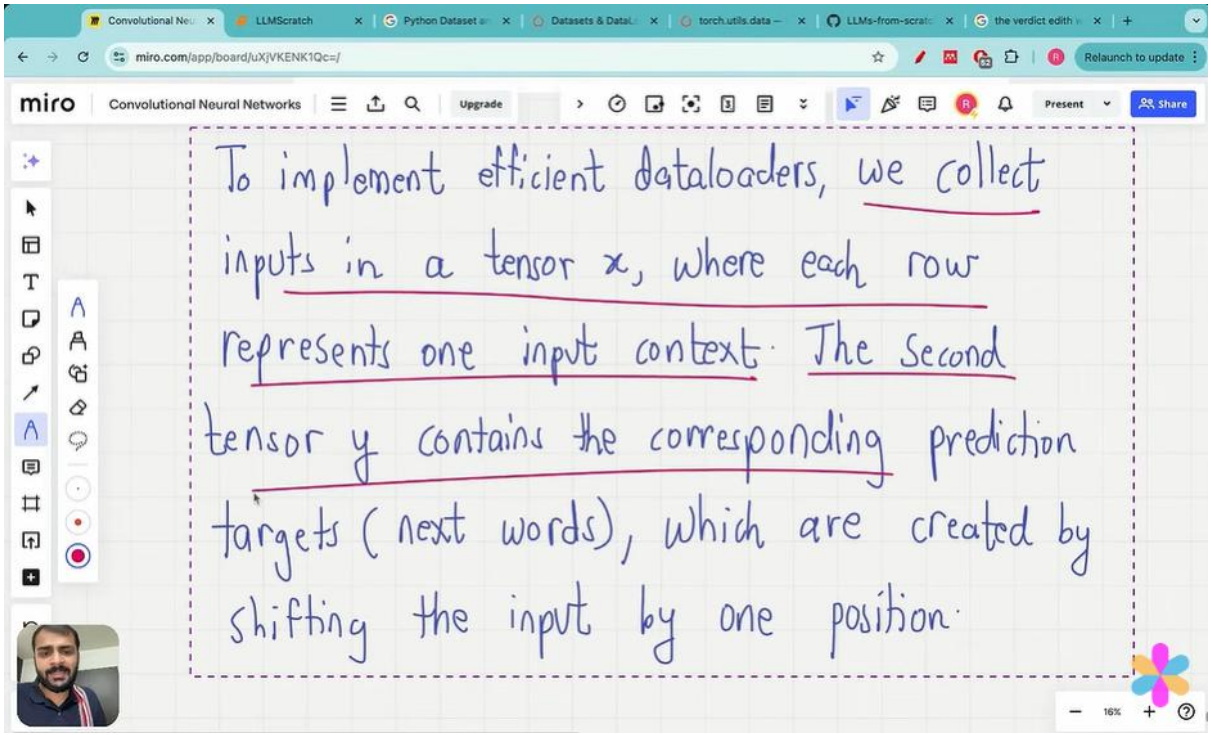

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LLMs learn to predict one word at a time
Given a text sample
extract input blocks as subsamples that serve as input to the LLM.
The LLMs prediction task during training is to predict the next

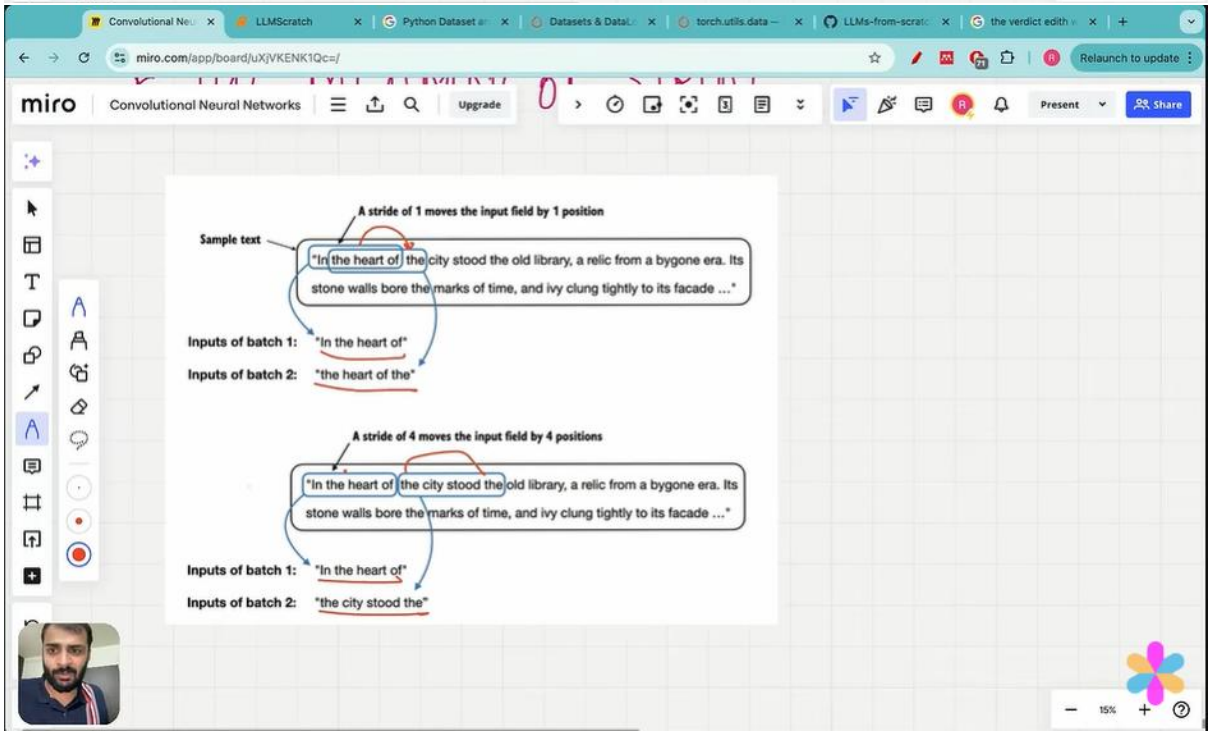
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training is to predict the next word that follows the input block.
During training, we mask out all words that are past the target.
* We will implement a data loader that fetches input-output target pairs, using


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* CREATING A DATA LOADER
Sample text
"In the heart of the city stood the old library, a relic from a bygone era. Its stone walls bore the marks of time, and ivy clung tightly to its facade ..."
Tensor containing the inputs
x = tensor([["In", "the", "heart", "of"],
["the", "city", "stood", "the"],
["old", "library", ",", "a"],
[_]])
Tensor containing the targets
y = tensor([["the", "heart", "of", "the"],
["city", "stood", "the", "old"],
["library", "a", "relic", "from"],
[_]])
To implement efficient dataloaders, we collect

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Convolutional Neural Networks
To implement efficient dataloaders, we collect inputs in a tensor x, where each row represents one input context. The second tensor y contains the corresponding prediction targets (next words), which are created by shifting the input by one position.


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Convolutional Neural Networks
A stride of 1 moves the input field by 1 position
Sample text
"In the heart of the city stood the old library, a relic from a bygone era. Its stone walls bore the marks of time, and ivy clung tightly to its facade ..."
Inputs of batch 1: "In the heart of"
Inputs of batch 2: "the heart of the"
A stride of 4 moves the input field by 4 positions
"In the heart of the city stood the old library, a relic from a bygone era. Its stone walls bore the marks of time, and ivy clung tightly to its facade ..."
Inputs of batch 1: "In the heart of"
Inputs of batch 2: "the city stood the"